As an example, we present

Proposition 5.13 (E73 $\not\models$ E4380)**.** *The law* $\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{y}))$ (E73) *does not imply* $\mathrm{x} \diamond (\mathrm{x} \diamond \mathrm{x}) \simeq (\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x}$ (E4380).

Proof. To build a rule set Γ that will imply E73 when total, a natural first choice would be the single rule

1. If $y \diamond (x \diamond y)$ is defined, then $y \diamond (y \diamond (x \diamond y))$ is defined and equal to x.

However, the greedy algorithm will fail just with this rule: if the partial operation has $x \diamond y$ and $z \diamond y$ both equal to some w for some $x \neq z$, then any attempt to assign a value to $y \diamond (y \diamond w)$ will lead to a contradiction, as the above rule will force $y \diamond w$ to equal both x and z. Indeed, it is clear that E73 forces all the right translation operators R_y to be injective. We therefore add this as an additional rule:

2. If $x \diamond y$ and $z \diamond y$ are defined and equal, then $x = z$.

To avoid some unwanted edge cases, it is also convenient to impose the additional rule

3. If $x \diamond y$ is defined, it is not equal to y.

Unlike Rule 2, this rule is not forced by E73, but can be enforced as part of the greedy construction.

The ruleset clearly satisfies the soundness axiom (ii) of Theorem 5.12. We now verify the greedy extension axiom (iii). Let Ω, a, b be as in that axiom. We may assume that $a \diamond b$ is undefined, since we are done otherwise. We adjoin a new element c to M to create M', and set $a \diamond' b = c$. If we also have $b = d \diamond a$ for some d (unique by Rule 2, and only possible for $a \neq b$ by Rule 3), set $a \diamond' c = d$ (this is necessary to retain Rule 1). Of course, we also set $x \diamond' y = x \diamond y$ whenever $x \diamond y$ is already defined.

Since $c \notin M$, it is clear that $\diamond'$ is a finitely supported partial magma operation on M'. It is also clear that $\diamond'$ satisfies Rule 2 and Rule 3. Now we case check Rule 1:

- Case 1: $x = c$ or $y = c$. Not possible since no left multiplication with c is defined.
- Case 2: $x \diamond' y = c$. Only possible when $x = a$, $y = b$, but then $y \diamond' (x \diamond' y)$ is undefined since $y = b \neq a$ if d is defined.
- Case 3. $y \diamond' (x \diamond' y) = c$. Only possible when $y = a$ and $x = d$, and holds in this case.
- Case 4: $x, y, x \diamond' y, y \diamond' (x \diamond' y) \neq c$: In this case $\diamond' = \diamond$ on all pairs, so the claim reduces to Rule 1 for $\diamond$, which holds by the induction hypothesis.

To conclude, we need to locate a seed $\diamond_0$ satisfying Rules 1,2,3 and containing a counterexample to E4380. One simple example is the carrier $\{0, 1, 2, 3\}$ with $0 \diamond_0 0 = 1$, $0 \diamond_0 1 = 2$, $0 \diamond_0 2 = 0$, $1 \diamond_0 0 = 3$. $\square$

This method is not guaranteed to halt in finite time, as there may be increasingly lengthy sets of rules one has to add to Γ to avoid collisions. However, in practice we have found many of the refutations that could not be resolved by simpler techniques to be amenable to this method (or variants thereof, as discussed below).

One can automate the above procedure by using ATPs (or SAT solvers) to locate new rules that are necessary and sufficient to resolve any potential collision (and which, *a posteriori*, can be seen to be necessarily consequences of the law E). The seed-finding step (Step 3) is particularly easy to automate, and can also often be done by hand. In some cases, the SAT solver calculations provided by these methods were difficult to formalize efficiently in Lean, and so we elected in some cases to replace computer-generated rulesets with shorter human-generated versions in preparation for the formalization step.

However, in some cases we have found it necessary to add "inspired" choices of rules that were not forced by the initial hypothesis E, but which simplified the analysis by removing problematic classes of collisions from consideration. We were unable to fully automate the process of guessing such choices; however, we found ATPs very useful for testing any proposed such guess. In particular, if an ATP was able to show that the existing ruleset, together with a proposed new rule A, implied E$'$, then this clearly indicated that one should not add A to the rule set Γ. Conversely, if an ATP failed to establish such an implication, this was evidence that this was a "safe" rule to impose.

We also found that human verification of the greedy extension property was a highly error-prone process, as the case analysis often included many delicate edge cases that were easy to overlook. Both ATPs and the Lean formalization therefore played a crucial role in verifying the human-written greedy arguments, often revealing important gaps in those arguments that required either minor or major revisions to the rule set.

The greedy method can also be combined with the translation-invariant method, both in abelian and non-abelian settings. For instance, we can modify the proof of Theorem 5.12 to obtain the following variant:

Theorem 5.14 (Noncommutative translation-invariant greedy algorithm)**.** *Let F, F' be functional equations on groups, and let Γ be a theory of first-order sentences regarding a partial function $f\colon \Omega \to G$ on a group $(G, \cdot, \cdot^{-1}, 1)$. Assume the following axioms:*

(i) (Seed) There exists a finitely supported partial function $f_0\colon \Omega_0 \to G$ satisfying Γ that contradicts F', in the sense that there is some assignment of variables in F' in G such that both sides of F' are defined using f_0, but not equal to each other.

(ii) (Soundness) If $f_n\colon \Omega_n \to G$ is a sequence of partial functions satisfying Γ with each f_{n+1} an extension of f_n, and the direct limit f_∞ is total, then this limit satisfies F.

(iii) (Greedy extension) If $f\colon \Omega \to G$ is a finitely supported partial function satisfying Γ, and $a \in G$, then there exists a finitely supported extension $f'\colon \Omega' \to G'$ of f to a possibly larger group G', and also satisfying Γ, such that $f'(a)$ is defined.

Then $F \not\models F'$.

One can of course also develop an abelian analogue of the above theorem, in which $(G, +, -, 0)$ and $(G', +, -, 0)$ are now required to be abelian. We can then give an alternate proof of Proposition 5.13 as follows:

Second proof of Proposition 5.13. (Sketch) The functional equations associated to E73 and E4380 are $f^2(h^{-1}f(h)) = h^{-1}$ and $f^2(1) = f(1)f(f(1)^{-1})$ respectively. We apply Theorem 5.14 with the following ruleset:

1. If $f(h^{-1}f(h))$ is defined, then $f^2(h^{-1}f(h))$ is defined and equal to h^{-1}.
2. If $h^{-1}f(h)$ and $k^{-1}f(k)$ are defined and equal, then $h = k$.
3. If $f(h)$ is defined, it is not equal to h.

Axiom (ii) is clear. To verify axiom (iii), we can assume $f(h)$ is undefined, then adjoin an element c freely to G to create a larger group G', and set $f'(h) = c$. If $h = k^{-1}f(k)$ for some k (which is unique by Rule 2, and only possible for $h \neq 1$ by Rule 3), then also set $f'(c) = k^{-1}$. One can then check that axiom (iii) is satisfied. For axiom (i), take G to be a free cyclic group with one generator a, and set $f(1) = a$, $f(a) = a^3$, $f(a^3) = 1$, $f(a^{-1}) = a^3$ (say). □

More complex (and *ad hoc*) variants of the greedy algorithm are possible. In some cases, we were not able to preserve the finitely supported nature of the partial operation or partial function, and needed to extend that partial object at an infinite number of values at each step. In other cases, one also had to add additional temporary data during the greedy process to record tasks that one wished to attend to at a later stage of the process, but could not handle immediately because it was awaiting some other operation to become well-defined. We will not attempt to survey all possible variants of this method here, but refer the reader to the ETP blueprint for further examples.

5.6. **Modifying base models.** A general technique that we have found useful in obtaining a refutation such as $\mathrm{E} \not\models \mathrm{E}'$ is to start with a simple base model $\mathcal{M} = (M, \diamond)$ that satisfies both E and E′, and modify it in various ways to preserve E, but create a violation of E′. There are many such possible modifications, but three general ways that have proven effective are as follows:

(i) Modify the magma operation $\diamond\colon M \times M \to M$ on a small set in order to violate E′, and then make further modifications as needed to recover E.
(ii) Construct an *extension* $\mathcal{N}$ of $\mathcal{M}$, equipped with a surjective magma homomorphism $\pi : \mathcal{N} \to \mathcal{M}$, and defined in terms of some additional data. Then solve for that data in such a way that $\mathcal{N}$ satisfies E but not E′.
(iii) Construct an *enlargement* $\mathcal{M}' = (M', \diamond')$ of $\mathcal{M} = (M, \diamond)$, which is a magma that contains $\mathcal{M}$ as a submagma. One needs to construct the multiplication table $\diamond$ on $(M' \times M')\backslash(M \times M)$ in order to retain E but disprove E′.

One appealing case of (ii) that our project discovered, involving a "magma cohomology" analogous to (abelian) group cohomology, is that of an *affine* extension of a magma $\mathcal{G} = (G, \diamond_G)$ by another magma $(M, \diamond_M)$ which is an abelian group M with a linear magma operation $s \diamond_M t := as + bt$ for some endomorphisms $a, b \in \mathrm{End}(M)$. One can then consider extensions with carrier $G \times M$ and magma operation

$$(x, s) \diamond (y, t) := (x \diamond_G y, s \diamond_M t + f(x, y)) \tag{8}$$